

WHITEPAPER

HOW BLOCKCHAIN COULD CHANGE THE FINANCIAL MARKETS

The New Kid on the Block



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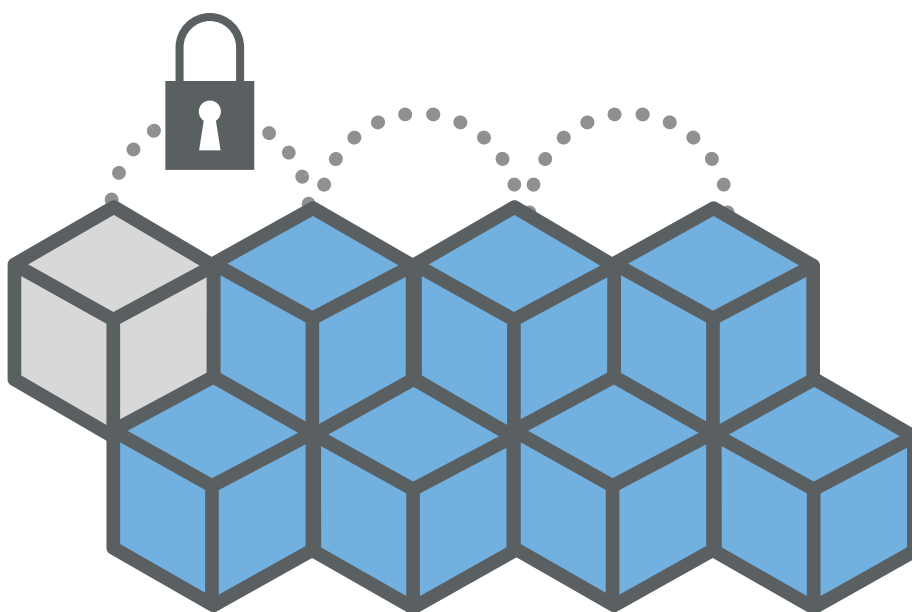
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Executive Summary

One of the most talked about topics in the financial markets today is blockchain. The global financial markets are investing in blockchain technology to revolutionise many aspects of financial products and services. Blockchain, sometimes referred to as Internet 3.0 or the Internet of Value, follows a strong growth trajectory and has quickly become a seamless part of the global financial system and economy. The blockchain technology platform is a distributed ledger technology (DLT) system, which has triggered a fundamental challenge to the nature of money, transforming current business processes. It is one of the most disruptive technologies available at present, designed to simplify the value chains around trading, payment and market infrastructure. If fully adopted, blockchain will create a more efficient, more transparent and more secure marketplace whilst reducing transaction processing costs.

Whilst the potential is huge, so too is the uncertainty. The key to turning blockchain's potential into reality is a collective effort amongst industry participants to learn, share, cooperate and see themselves as part of the blockchain network, as opposed to individual firms.

This short primer focuses on the applications of blockchain technology on various aspects of the capital markets. It should be read in conjunction with 'Blockchain Technologies in Financial Markets' and builds upon the technical concepts covered there.



Financial Applications of Distributed Ledger Technology

The potential applications of blockchain and Distributed Ledger Technology to the world of finance (beyond the headline grabbing hype of the cryptocurrency roller-coaster and the more recent controversial growth of fundraising Initial Coin Offerings) has been an area of growing interest and research. Banks, payment processors, financial software companies, regulators and governments are actively investigating and developing the necessary technology to address some of the ingrained structural and operational challenges as well as the inefficiencies of modern finance.

There is plenty to write and discuss about the potential impact of blockchain on cross-boarder payments, generalised digital tokenisation of value, remittances and the lowering of barriers for access to finance in third world countries. This paper will focus on two important areas for capital markets – blockchain's impact on financial transactions (OTC derivatives and syndicated loans) and risk management.

The OTC Derivatives Industry

According to BIS [1], the Over-the-Counter (OTC) derivatives market, with an outstanding notional of \$542 trillion as of 2017, is a highly intermediated marketplace consisting of buyers, sellers, broker-dealers and Central Counterparty Clearing houses (CCPs).

The trade life-cycle consists of three main stages:

- 1 Pre-Trade:** Buy and sell orders are linked by broker-dealer desks or execution facilities (swaps). OTC contracts are bilateral instruments with standard or bespoke terms.
- 2 Trading / Clearing:** Once a buyer and seller have been matched, the trade is either executed bilaterally (non-cleared) or cleared via a Central Counterparty Clearing House (CCP). In the CCP clearing framework, CCP acts as the counterparty for both 'legs' of the transaction via the process of novation. Counterparty risk of non-CCP cleared transactions is mitigated through netting and/or collateralisation. However, CCP-cleared trades must meet strict margin requirements set by the CCP, which further hedges away tail risk through application of risk mutualisation via waterfall models and other hedging mechanisms. CCPs allows for significant counterparty risk mitigation via compression, netting and the maintenance of margin, whilst simultaneously creating massive novel systemic risk.
- 3 Settlement and Delivery:** Once the contract matures (or is triggered), it is closed through either cash settlement (typical) or physical delivery of the underlying. Settlement processes take time (depending on the type of contract), but usually 1-3 business days. During this period, any non-simultaneous exchange of payments constitutes settlement risk, whereby either side can fail to pay or deliver.

Based on the above trade life-cycle, there are several ways for DLTs to transform the derivatives landscape, specifically in trading, trade reconciliation and reporting.



Trade Reconciliation

Currently, any derivatives transaction consists of a contract between two (or more) parties, which may be evaluated via different models (pricing) as well as reported via different accounting procedures. Consensus between market participants is achieved via periodic reconciliation, or, if disputes occur during trade life-cycle events and cashflow exchanges, the involvement of other legal or trusted authorities. DLT offers the possibility of migrating to a whole new framework, whereby the terms of a derivatives transaction are captured in a single, mutually agreed upon and, thereafter, immutable smart contract. As the trade matures, events such as floating cashflows payments, contingent claims, etc. would be triggered via data fed from trusted sources (Oracles) with automatic settlement and transfer of funds being handled by the smart contract algorithm. As the valuation model is embedded into the contract, pricing and cashflow disputes are less likely to occur. Moreover, consensus is achieved in a distributed fashion with validators of the network (potentially Significant financial institutions [SIFIs], regulators, etc.) ensuring network robustness and quick dispute resolution (for example, via proof of authority). Smart contracts, as applied to derivatives trading and post trade processing, will improve efficiency and lower costs of capital by reducing reconciliation time and dispute processing in the financial markets [2].



Regulatory Reporting

All actions and events are stamped on the distributed ledger. A regulator running a full node would have complete access to the complete history of trading activity and payments across the network on a near real-time basis.



Auction Process

Notwithstanding the speed of confirmation on-chain, the process of order matching may be made simpler via a set of interoperable blockchains. OTC participants would submit bids directly to the network and rely on smart contracts to automatically choose the highest bid via a new form of decentralised exchange.

Progress on some of these financial applications is already underway. Last year, DTCC, in partnership with technology vendors, began working on a DLT solution for credit derivatives processing, which will enable the management of post-trade events for CDS trades on a distributed, permissioned DLT network [3]. Even more ambitious in scope and impact is the Common Domain Model (CDM), which is being developed by ISDA in conjunction with REGnosys [4,5]. The aim is to provide a global representative standard for all events and actions occurring during the life of a trade as well as the capture and automation of all associated product data (via the well-established FpML framework) onto a smart contract blockchain. ISDA recognised that automation and interoperability can only succeed when there is standardisation, and their long-term vision is to establish the CDM as the market standard to reduce reconciliation overheads and reporting inconsistencies (see Case Study: ISDA Common Domain Model on page 10).

Broadly Syndicated Loans (BSL)

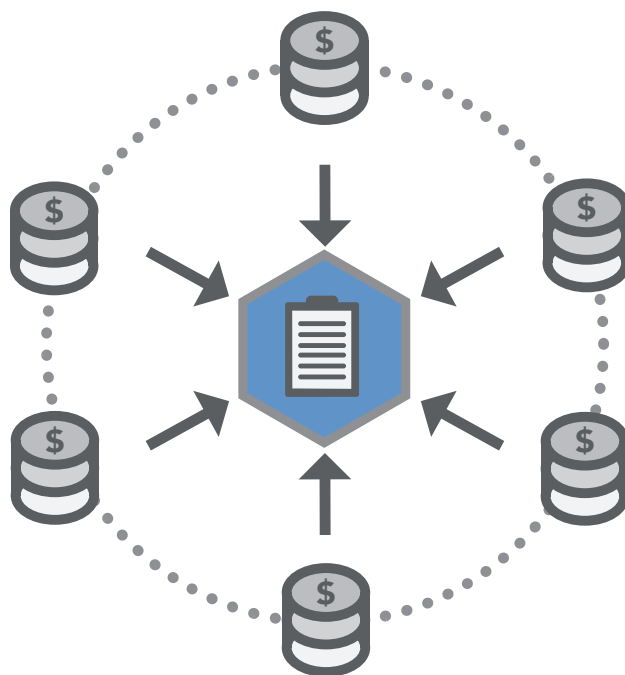
Syndicated loans provide borrowers with a means of securing a loan facility from a group of lenders through a single loan agreement. Since such an agreement applies to the whole group of lenders, it eliminates the need for multiple, separate bilateral loans documents, therefore, simplifying the borrowing process. The syndicated loan space is rather complicated. For instance, loans can be structured as different tranches, and loan facilities can be a revolving credit line (e.g. a letter of credit, bridge loans and amortizing term loans).

An agent bank is appointed by the lenders to administer the loan. Its role can be extremely intensive due to the ongoing monitoring of the various facilities. Here we illustrate few key responsibilities:

- Representing the syndicate and providing a single point of contact for the borrower
- Monitoring compliance of the borrower with key terms of the facility
- Principal, interest, fee disbursements and reconciliations
- Maintenance of register and processing of assignments

Distributed ledger technologies can potentially automate the agent bank role via a single source of truth to achieve down-stream automation of various administrative tasks. Further, DLT could provide full visibility on the chain of ownership of a loan, thus reducing the friction around reps and warranties – in particular, when a loan goes from par to distressed status and lawyers may be required.

Like most fixed income products, BSLs, in recent years, have grown in size and the structure of ownership has changed to include participation by mutual funds. Regulators and the fixed income community support improvement in secondary market liquidity and a fund's ability to assess the liquidity of its underlying holdings. We believe DTL can be a valuable tool to remove operational latency from this market.



Risk Management after the Global Financial Crisis of 2008

The Global Financial Crisis (GFC) of 2008 was precipitated by the chain default of subprime mortgages in the US. This triggered significant mark-to-market write-downs and impairments on derivative instruments based on the securitisation of those mortgages (CDOs) and led to massive losses of capital across the global banking system.

Bitcoin was a direct response to GFC and the subsequent QE programme. Its genesis block contained the now famous headline from The Times UK on the 3rd Jan 2009 [6]: *Chancellor on brink of second bailout for banks*.

The crisis exposed the frail, interconnected nature of the global markets and their dependency on 'too big to fail' entities as well as the ability to devalue money at will by Central Banks.

The financial crisis highlighted two key types of risk associated with the derivatives market. The first is counterparty risk, which is a measure of the loss incurred in the event of a default of a counterparty prior to the maturity of the contract (pre-settlement risk) or settlement of any due payments (settlement risk). The second risk is systemic risk associated with key institutional failures, leading to the spread of contagion that triggers the collapse of other institutions (as illustrated by Bear Stearns and Lehman Brothers).

Learning from the lessons of 2008, the current regulatory framework, as promulgated under Basel III [7] (and recent amendments), places more stringent requirements on the amount of capital that banks must hold in reserve as well as seeking to deleverage balance sheets (via the Leverage Ratio) and improve provisions for liquidity in times of stress (Liquidity Coverage Ratio and Net Stable Funding Ratio) [8]. The market implementation of these requirements has led to an industry-wide improvement in the measurement and modelling of counterparty credit exposure (SACCR and Monte Carlo simulations) as well as a drive towards the minimisation of such exposures via netting and collateralisation.

Netting is the process of aggregating a set of trade level exposures between trading counterparties, such that positive and negative exposures can offset each other to arrive at a reduced total exposure.

Another risk mitigation approach is collateralisation, whereby both counterparties periodically reconcile their netted exposure throughout the lifecycle of their set of transactions and make payments to each other on any amount that is owed (via cash or other assets subjected to a haircut). Should either party default prior to maturity, their counterparty will have recourse to the collateral that they had received up until that point (known as Variation Margin [VM]). Upon default, there is typically a delay between the last collateral update and the final liquidation of the defaulting party's assets, known as the Margin Period of Risk (MPOR). This can be anywhere between 3 to 40 days, or more, depending on disputes, litigation and other bottlenecks within the liquidation process. This introduces significant Gap Risk, since the value of defaulted transactions will likely have deviated significantly since the last collateral update. To cover this, an additional amount of collateral known as Initial Margin (IM) is needed (typically determined via Value-at-Risk, or more recently, sensitivities under the ISDA Standard Initial Margin Model [SIMM]) [9]. CCPs impose IM and VM by default, whilst non-cleared OTC derivatives can be collateralised via CSAs. Since 2017, regulators have started to phase in mandatory margin requirements for non-cleared trades in an attempt to incentivise CCP clearing (and more standardised contracts).

Blockchain-Enabled Risk Management Framework

Distributed ledger technology presents a new paradigm for how information is collated and distributed and is poised to revolutionise the way firms transact. It mitigates risk and addresses settlement, pre-settlement risk and collateral management practice.

Managing Settlement and Counterparty Risk on the Blockchain

Settlement risk is prevalent in the non-cleared OTC Derivatives market as there is no CCP to guarantee payment. Blockchain eliminates settlement risk, insofar as smart contracts can be designed to ensure the simultaneous exchange of payments, principal and underlying, upon contract maturity.

Blockchain technology has the potential to transform the management of pre-settlement risk. For instance, the process of collateralisation for non-cleared OTC derivatives can be automated via smart contracts. These contracts can be used to implement Collateral Support Annex (CSA) terms (payment rules, thresholds, minimum transfer amounts, etc.) and the valuation model required to calculate the variation margin. By embedding CSA terms into smart contracts, we can reduce disputes over collateral payments for the set of transactions [10].

Automation on the blockchain can increase the collateral monitoring frequency from daily (current best case) to hourly or even pseudo real-time. The close out process itself can also be transformed, since a missed variation margin payment can be flagged in a matter of hours or minutes, rather than days, and thus, the effective MPOR is reduced. A shorter MPOR results in a significant reduction in initial margin that needs to be held in reserve, freeing up capital and lowering Capital Valuation Adjustment (KVA) charges.



In a smart contract framework, we can programmatically add grace periods or wait for a consecutive number of such defaults to occur before automatically unwinding the positions, therefore reducing the chances of technical defaults due to liquidity gaps.

Managing Collateral on the Blockchain

The process of collateral management and lending can also be automated on the blockchain as the distributed ledger enables complete tracking of asset ownership and transaction history.

Assets held as collateral (both cash and non-cash) in a variety of custodian accounts can be recorded on the ledger. A smart contract automatically evaluates current asset prices based on predefined haircuts and market data (interest rates, FX rates, equity prices, etc.) supplied by Oracles. Once the collateral value falls below a required amount, the contract triggers collateral calls or unwinds the positions.

under the contract. Similar types of smart contract have been designed to create cryptocurrencies known as stable coins. The intrinsic value of these currencies is pegged to other fiats such as USD, but backed by tokenised assets, such as Ethereum, via Collateralised Debt Positions (e.g. DAI by MakerDAO) [11].

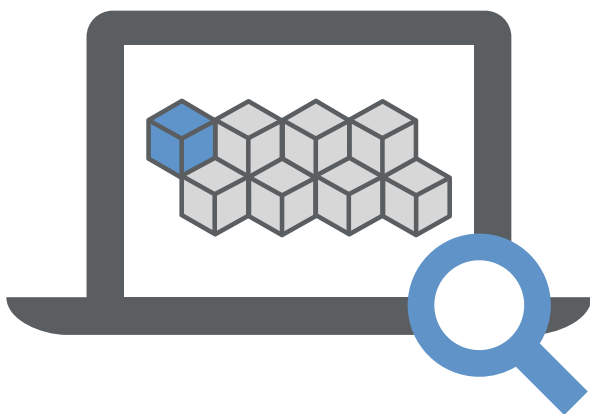
In traditional banking, collateral lending or re-hypothecation mitigates liquidity and funding costs associated with over-collateralisation, whereby excess collateral can be lent or used as High Quality Liquid Assets (HQLAs) for other purposes. The main drawback of this approach is that it involves a physical transfer of assets between accounts and leads to additional credit risk and repayment risk. A consortium of banks including Commerzbank, Credit Suisse and UBS, amongst others, have developed a blockchain solution for a collateral lending framework in the form of Digital Collateral Receipts (DCR) using the Corda DLT [12]. The DCR is a digital token that enables the exchange or legal ownership of a basket of securities, without the need to migrate underlying assets across custodians. The tokens can then be used to pledge such securities for IM and VM as well as to provide HQLA in order to meet LCR and NSFR liquidity requirements. Conditional logic in the smart contract can enforce the creator of the token to maintain the notional value above a pre-defined

threshold or automated liquidation protocols when funds are no longer sufficient. Finally, the DCR network enables regulators to have a real-time view of collateral allocation and usage, allowing potential liquidity and funding shortfalls to be identified in a timely and efficient manner.

Managing Systemic Risk on the Blockchain

The rise in prominence of CCPs, post-2008, as a means of reducing counterparty risk has, to a certain extent, introduced additional systemic and contagion risk into the global financial markets. Risk management of CCPs continues to be an active area of research and development with the focus on large global regulatory bodies.

Blockchain proponents argue that the decentralised nature of the technology could mitigate the risk of massive single point failures, as represented by the current framework. Rather than CCPs, a decentralised clearing network (DCN), organised into a distributed autonomous organisation (DAO), could fulfil some key functions of current CCPs, such as contract and margin valuations, collateralisation and closeout management as well as novation, netting and even compression [2]. Validators of such a network (or a pool of interoperable blockchain networks) would need to meet certain admission criteria (equivalent to CCP membership requirements) and serve to safeguard the network by providing capital, punishing bad actors and providing consensus (either via Proof of Stake, or Proof of Authority). Moreover, all participants of the DCN, in contrast to the current unilateral approach of the CCP, can vote for changes to the rules (such as margin requirements and capital buffers). Such blockchain infrastructure would be open and transparent to regulatory oversight, paving the way for a fairer, more efficient and robust derivatives market.



Case Study: ISDA Common Domain Model

The International Swap Dealers Association (ISDA) has been leading efforts to standardise the modeling of financial instruments for many years, including the development of Financial products Markup Language (FpML) in 1999. FpML was developed to standardise the communication of complex derivatives contract terms between counterparties and to simplify messaging of trade data. Even with standardised messaging, firms still use individual processes to model, process and value complex derivatives. To further streamline the post-execution trade cycle, ISDA Common Data Model (CDM) focuses on common, industry-wide features of the trade life-cycle. It serves as a data model specification and “is targeted at DLTs to exploit their embedded lineage and consistency properties.”

The CDM infrastructure set forth the foundations for DLTs to automate data and to synchronise the state of derivatives contracts across various trading counterparties. DLT technologies allow for a consistent record to be shared among all relevant market participants simultaneously. Further, it can also contain trade life-cycle events as well as risk and valuation information, therefore transforming data integrity and reducing reconciliation and reporting costs for the financial services industry.

Designed to facilitate the instantiation of trades on a DLT, CDM is the first step to building self-executing smart contracts. Rather than beginning with the product, such as a 5yr USD Interest Rate Swap (IRS), the CDM begins at the “event” level. An event represents an action that may be applicable to any trade e.g. initiating, cancelling, amending, novating the trade, etc. The events of the first stage identify common actions, which are independent of the financial products; the second layers are “dependent events” defined by contract terms, market data, record of daily value, option exercise, etc. For example, a simple Forex trade is an event, namely, the exchange of cash between two parties, and the price is captured by the two amounts exchanged. We can build a record of more complex transactions from simple derivatives that are built from independent and dependent events. The process scales up to whole portfolios, allowing for the automatic operation of portfolio events such as collateral calls, capital and risk calculations.

Example: Axoni's Implementation of ISDA Common Domain Model

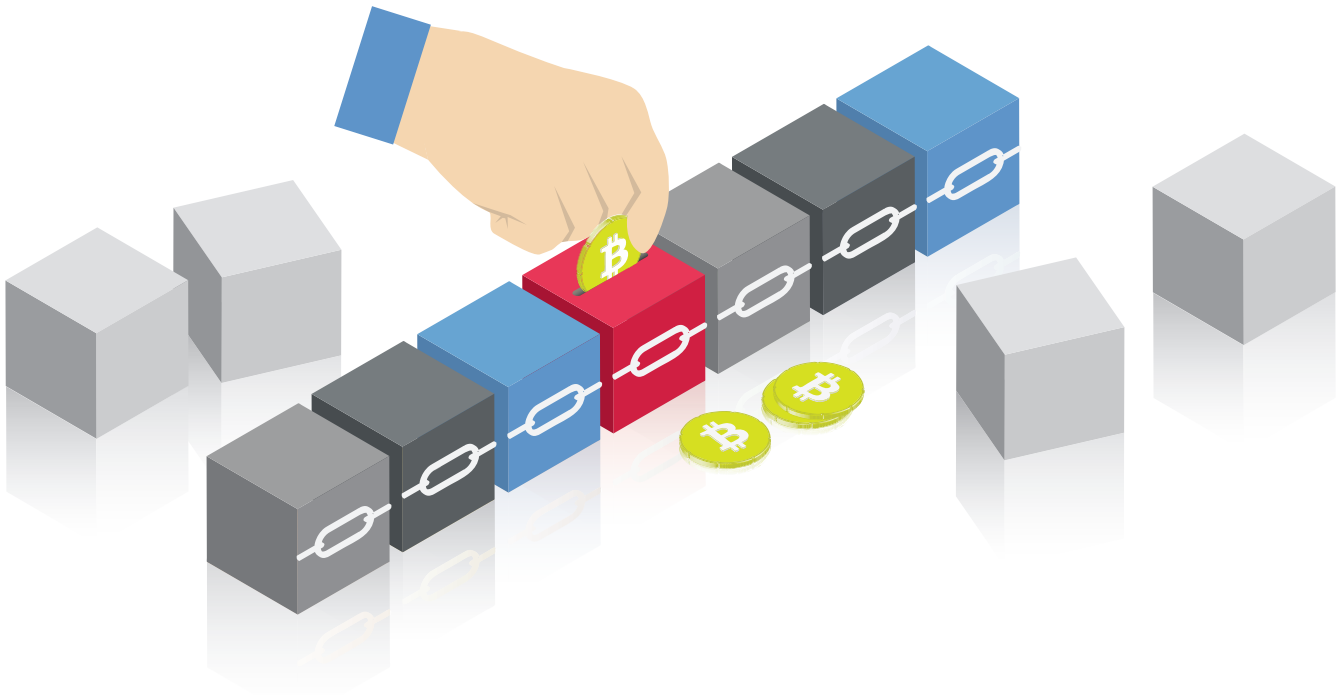
Axoni has developed a permissioned distributed ledger technology infrastructure to implement the ISDA CDM based on Scala programming language (AxLang). In a blockchain network, each party is executing the same code in the same deterministic environment. To capture the benefits of blockchain technology's synchronicity and cross-system automation, whilst minimising the risk of such errors, extra precautions are applied to ensure smart contracts are error-free.

AxLang smart contracts employ a mathematical, formal verification approach to prove the correctness of computer programs. It supports

both public and private Ethereum projects and leverages upon existing Scala user base and compile to either the Java Virtual Machine (JVM) or to the Ethereum Virtual Machine (EVM). It can be used to write formally verified smart contracts on a distributed ledger or to run an off-ledger locally in a standard Java environment, providing flexibility to test under various environments with ease.

To test the principles of the ISDA CDM, Axoni have recently implemented and deployed CDM into an infrastructure that enables full state synchronisation between parties.

Closing Thoughts



Blockchain (or distributed ledger) technology has generated a huge amount of interest within financial markets. Whilst the business benefits of blockchain are clear, the industry needs to collaborate with government agencies and regulators to realise the potential of blockchain.

The shift and adoption of a new technology standard will take time. The regulator is one of the most important stakeholders in the adoption of this new technology, as new regulatory principles are likely to be needed where blockchain technology becomes an integral part of the market infrastructure. Some regulators have expressed interest as they recognise the potential impact on business models, reductions in risk and savings of cost and capital. However, it is hard to predict when and where DL technology applications will reach scale and what kind impact they will have across markets. For firms that want to benefit from DL technology, an important first step is to understand the scope of its likely impact on the market and to establish a clear view of the financial, technology and regulatory ecosystem.

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